

Flux Limiting or Extremum Regularization? : A Preliminary Study in Conservative Physics-Informed Neural Networks

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Abstract

Conservative physics-informed neural networks provide a domain-decomposed framework for approximating solutions of conservation laws, but their smooth neural representation can have difficulty near shock discontinuities. This study examines a minmod-inspired local-extremum penalty for a cPINN approximation of the one-dimensional isentropic Euler equations. Unlike a classical finite-volume flux limiter, the proposed term does not modify reconstructed interface states or numerical fluxes; it instead penalizes changes in the signs of neighboring discrete differences of the neural prediction. Preliminary experiments show that the baseline cPINN and the regularized cPINN both reproduce the principal wave structure of a high-resolution finite-volume reference solution. In the initial experiment, however, the regularized model does not improve the reported error, motivating a systematic study of the penalty weight, smooth-extremum suppression, total variation, shock location, and sensitivity to random initialization. The work therefore investigates the distinction between conservative flux limiting and limiter-inspired neural regularization rather than claiming that the two procedures are equivalent.

keywords Computational Fluid Dynamics, Physics-Informed Neural Networks, Numerical Schemes, Applied Mathematics.

1. INTRODUCTION

Hyperbolic conservation laws arise in many models of compressible flow and can develop shock discontinuities even when their initial data are smooth. High-resolution finite-volume methods commonly use flux or slope limiters to suppress nonphysical oscillations near discontinuities while retaining higher-order accuracy in smooth regions. In a classical Monotonic Upwind Scheme for Conservation Laws (MUSCL) type method, the limiter acts on reconstructed slopes and thereby modifies the left and right states supplied to a numerical flux. This mechanism is fundamentally different from adding a regularization term to the loss function of a neural-network solver.

Physics-informed neural networks (PINNs) approximate the solution of a partial differential equation by minimizing losses associated with the governing equations, initial and boundary data, and other physical constraints [1]. In the presence of shocks, however, a smooth neural-network representation may have difficulty minimizing a pointwise strong-form residual because the physically relevant solution satisfies the conservation law in a weak sense at a discontinuity.

This study uses the conservative physics-informed neural network (cPINN) framework introduced by Jagtap et al. [2]. A cPINN partitions the spatial domain into multiple nonoverlapping subdomains and assigns a separate neural network to each subdomain. The networks are coupled by losses that enforce consistency of the predicted states and physical fluxes at shared interfaces. This domain decomposition can facilitate parallel training and allows different parts of the solution to be represented

by separate networks. Nevertheless, interface-flux consistency in a cPINN is not equivalent to a cellwise finite-volume balance, and the term “conservative” should not be interpreted as meaning that the cPINN is itself a finite-volume method.

The purpose of the present study is to investigate a minmod-inspired local-extremum penalty added to the cPINN training objective. For neighboring neural predictions U_{i-1} , U_i , and U_{i+1} on a structured evaluation grid, the penalty activates when the backward and forward differences have opposite signs. It therefore discourages strict discrete local extrema. Importantly, this term does not reconstruct interface states and does not modify a numerical flux. It is consequently described in this paper as an *extremum regularizer*, rather than as a neural flux limiter.

The central research question is whether this extremum regularizer improves the approximation of shock-containing solutions or whether it suppresses valid solution features and increases error. We compare a baseline cPINN with an extremum-regularized cPINN and evaluate both against a high-resolution finite-volume reference solution. The available results are preliminary: they establish qualitative agreement between the cPINN predictions and the principal wave structure of the numerical reference, while a systematic penalty-weight and repeated-seed study remains in progress.

1.1. One-Dimensional Isentropic Euler Equations

We consider the one-dimensional isentropic Euler equations in conservative variables. Let $\rho(x, t) > 0$ denote density, let $u(x, t)$ denote velocity, and define the momentum by

$$m(x, t) = \rho(x, t)u(x, t).$$

The governing equations are

$$\frac{\partial \rho}{\partial t} + \frac{\partial m}{\partial x} = 0, \quad (1)$$

$$\frac{\partial m}{\partial t} + \frac{\partial}{\partial x} \left(\frac{m^2}{\rho} + p(\rho) \right) = 0. \quad (2)$$

The isentropic pressure law is

$$p(\rho) = K\rho^\gamma, \quad (3)$$

where $K > 0$ is the pressure constant and $\gamma > 1$ is the isentropic exponent.

Defining

$$U = \begin{pmatrix} \rho \\ m \end{pmatrix}, \quad F(U) = \begin{pmatrix} m \\ \frac{m^2}{\rho} + K\rho^\gamma \end{pmatrix}, \quad (4)$$

the system can be written in conservative form as

$$\partial_t U + \partial_x F(U) = 0. \quad (5)$$

Although the problem has only one spatial dimension, it contains two coupled conservation laws and two characteristic wave families. It is therefore more demanding than a scalar model such as Burgers’ equation. The isentropic Euler system describes compressible flow; it should not be described as a model of incompressible liquid flow.

The numerical experiments use a Riemann initial condition of the form

$$U(x, 0) = \begin{cases} U_L = (\rho_L, m_L)^\top, & x < x_0, \\ U_R = (\rho_R, m_R)^\top, & x > x_0. \end{cases} \quad (6)$$

In the experiments, $U_L = (1.0, 0.0)^\top$, $U_R = (0.125, 0.0)^\top$, $x_0 = 0.5$, $K = 1.0$, $\gamma = 1.4$, and the final time is $T = 0.15$. Zero-gradient (outflow) boundary conditions are used; see Section 2.1.

1.2. Lax–Wendroff Baseline

The Richtmyer form of the Lax–Wendroff method [3] is used as an unlimited finite-difference baseline for

$$\partial_t U + \partial_x F(U) = 0.$$

The predictor step evaluates an intermediate state at each cell interface:

$$U_{i+1/2}^{n+1/2} = \frac{1}{2} (U_i^n + U_{i+1}^n) - \frac{\Delta t}{2\Delta x} [F(U_{i+1}^n) - F(U_i^n)]. \quad (7)$$

The corrector step is

$$U_i^{n+1} = U_i^n - \frac{\Delta t}{\Delta x} [F(U_{i+1/2}^{n+1/2}) - F(U_{i-1/2}^{n+1/2})]. \quad (8)$$

For sufficiently smooth solutions, the method is second-order accurate in space and time. It is not monotonicity preserving, however, and may generate dispersive oscillations near discontinuities. In this study, Lax–Wendroff is treated as an unlimited baseline and is kept distinct from the limited MUSCL finite-volume method described below.

1.3. Classical MUSCL–Minmod Finite-Volume Method

To provide a classical limited comparison, we use a MUSCL reconstruction with minmod-limited slopes. For cell averages U_i , define

$$\sigma_i = \text{minmod} \left(\frac{U_i - U_{i-1}}{\Delta x}, \frac{U_{i+1} - U_i}{\Delta x} \right), \quad (9)$$

where the scalar minmod function is

$$\text{minmod}(a, b) = \begin{cases} \text{sgn}(a) \min(|a|, |b|), & ab > 0, \\ 0, & ab \leq 0. \end{cases} \quad (10)$$

For the system of equations, the manuscript must state whether minmod is applied componentwise or in characteristic variables.

The reconstructed interface states are

$$U_{i+1/2}^L = U_i + \frac{\Delta x}{2} \sigma_i, \quad (11)$$

$$U_{i+1/2}^R = U_{i+1} - \frac{\Delta x}{2} \sigma_{i+1}. \quad (12)$$

These states are supplied to the Rusanov numerical flux

$$\hat{F}_{i+1/2} = \frac{1}{2} [F(U_{i+1/2}^L) + F(U_{i+1/2}^R)] - \frac{a_{i+1/2}}{2} (U_{i+1/2}^R - U_{i+1/2}^L), \quad (13)$$

where $a_{i+1/2}$ is an estimate of the largest local characteristic speed. For the isentropic Euler equations, the characteristic speeds are

$$u - c \quad \text{and} \quad u + c, \quad (14)$$

where

$$u = \frac{m}{\rho}, \quad c = \sqrt{p'(\rho)} = \sqrt{K\gamma\rho^{\gamma-1}}. \quad (15)$$

This is the location at which classical minmod limiting acts: it changes the reconstructed states used to calculate the conservative interface flux. By contrast, the cPINN term studied in this work changes only the training loss and does not enter the Rusanov flux. The two mechanisms should therefore be compared, but they should not be described as mathematically equivalent.

2. METHODS

This section describes the finite-volume reference solution, the baseline conservative physics-informed neural network (cPINN), and the proposed extremum-regularized cPINN. The neural regularizer is not treated as a classical flux limiter because it does not reconstruct interface states or modify a numerical flux. Instead, it is added to the neural-network training objective and penalizes strict discrete extrema detected on a structured evaluation grid.

2.1. Computational Domain and Initial Data

The spatial domain is

$$x \in [0, 1],$$

and the solution is computed over

$$t \in [0, T],$$

where $T = 0.15$.

The initial condition is a Riemann problem of the form

$$U(x, 0) = \begin{cases} U_L = (\rho_L, m_L)^\top, & x < x_0, \\ U_R = (\rho_R, m_R)^\top, & x > x_0, \end{cases} \quad (16)$$

where

$$U_L = (\rho_L, m_L)^\top = (1.0, 0.0)^\top, \quad U_R = (\rho_R, m_R)^\top = (0.125, 0.0)^\top, \quad x_0 = 0.5.$$

The pressure-law parameters are

$$K = 1.0, \quad \gamma = 1.4$$

The boundary conditions are zero-gradient (outflow) conditions, imposed at each time step by copying the first and last interior cell values into the two edge cells:

$$\rho_0^{n+1} = \rho_1^{n+1}, \quad \rho_{N_x-1}^{n+1} = \rho_{N_x-2}^{n+1}, \quad (17)$$

$$m_0^{n+1} = m_1^{n+1}, \quad m_{N_x-1}^{n+1} = m_{N_x-2}^{n+1}. \quad (18)$$

All parameter values reported in this section must match the implementation used to generate the figures and error values.

2.2. Finite-Volume Reference Solution

A high-resolution finite-volume solution is used as a numerical reference for evaluating the neural-network predictions. This reference should not be described as an exact solution unless a separate exact or semi-analytic Riemann solver has been implemented.

The reference solution is computed using piecewise-linear MUSCL reconstruction with minmod-limited slopes and a Rusanov numerical flux [4]. For a cell average U_i , the limited slope is

$$\sigma_i = \text{minmod} \left(\frac{U_i - U_{i-1}}{\Delta x}, \frac{U_{i+1} - U_i}{\Delta x} \right), \quad (19)$$

where

$$\text{minmod}(a, b) = \begin{cases} \text{sgn}(a) \min(|a|, |b|), & ab > 0, \\ 0, & ab \leq 0. \end{cases} \quad (20)$$

The reconstructed states at the interface $x_{i+1/2}$ are

$$U_{i+1/2}^L = U_i + \frac{\Delta x}{2} \sigma_i, \quad (21)$$

$$U_{i+1/2}^R = U_{i+1} - \frac{\Delta x}{2} \sigma_{i+1}. \quad (22)$$

These states enter the Rusanov numerical flux

$$\hat{F}_{i+1/2} = \frac{1}{2} [F(U_{i+1/2}^L) + F(U_{i+1/2}^R)] - \frac{a_{i+1/2}}{2} (U_{i+1/2}^R - U_{i+1/2}^L), \quad (23)$$

where $a_{i+1/2}$ is an estimate of the largest characteristic speed at the interface. For the isentropic Euler system,

$$a_{i+1/2} = \max(|u_L| + c_L, |u_R| + c_R), \quad (24)$$

with

$$u = \frac{m}{\rho}, \quad c = \sqrt{K \gamma \rho^{\gamma-1}}.$$

The finite-volume update is

$$U_i^{n+1} = U_i^n - \frac{\Delta t}{\Delta x} (\hat{F}_{i+1/2} - \hat{F}_{i-1/2}). \quad (25)$$

The time integration method is The first-order explicit (forward Euler) time stepping applied directly to the Rusanov flux divergence — a single flux evaluation and update per step, with no intermediate RK stages. and the time step is selected using:

$$\Delta t = \text{CFL} \frac{\Delta x}{\max_i (|u_i| + c_i)}, \quad (26)$$

where

$$\text{CFL} = 0.4.$$

The reference calculation uses

$$N_x = 200$$

cells. The minmod limiter is applied componentwise, independently to ρ and m , rather than in characteristic variables.. The complete space–time history (ρ_i^n, m_i^n) is stored for reference and evaluation purposes.

2.3. Spatial Domain Decomposition

The cPINN spatial domain is divided into four nonoverlapping subdomains:

$$\Omega_1 = [0, 0.375], \quad (27)$$

$$\Omega_2 = [0.375, 0.5], \quad (28)$$

$$\Omega_3 = [0.5, 0.625], \quad (29)$$

$$\Omega_4 = [0.625, 1]. \quad (30)$$

Each subdomain is represented by an independent neural network, and neighboring networks are coupled through interface losses.

These subdomain boundaries are fixed for all baseline and regularized experiments. Using identical boundaries ensures that differences between the models cannot be attributed to a change in domain decomposition.

If the actual implementation selected the subdomain boundaries by examining the finite-volume reference history, the preceding paragraph must be replaced by an explicit description of that procedure. Such a model must then be described as *reference-assisted*, because information from the numerical reference is used to design the neural-network decomposition.

2.4. Neural-Network Architecture

For each subdomain Ω_s , $s = 1, \dots, 4$, a fully connected neural network

$$\mathcal{N}_s(x, t; \theta_s) \longmapsto (\rho_{\text{raw}}^{(s)}(x, t), m^{(s)}(x, t)) \quad (31)$$

is used to approximate density and momentum. The network has

- 4 hidden layers;
- 20 neurons per hidden layer;
- a locally hyperbolic-tangent activation;
- Xavier (truncated-normal) parameter initialization.

The inputs are mapped to normalized local coordinates within each subdomain. For example,

$$\hat{x} = 2 \frac{x - x_s^L}{x_s^R - x_s^L} - 1, \quad \hat{t} = 2 \frac{t}{T} - 1, \quad (32)$$

where x_s^L and x_s^R are the left and right boundaries of Ω_s .

Density positivity is enforced using

$$\rho^{(s)}(x, t) = \text{softplus}(\rho_{\text{raw}}^{(s)}(x, t)) + \epsilon_\rho, \quad (33)$$

where

$$\epsilon_\rho = 1 \times 10^{-6}.$$

This transformation prevents negative predicted densities and ensures that the pressure law $p(\rho) = K\rho^\gamma$ remains well defined. It does not, by itself, guarantee that the predicted solution satisfies an entropy condition.

2.5. Training Points

For each subdomain, the following training sets are constructed:

1. Initial-condition points

$$\mathcal{D}_{\text{IC}}^{(s)} = \{(x_j, 0, U_0(x_j))\}_{j=1}^{N_{\text{IC}}^{(s)}}.$$

2. Boundary-condition points on the external boundaries $x = 0$ and $x = 1$,

$$\mathcal{D}_{\text{BC}} = \{(x_{\text{bdy}}, t_j, U_{\text{bdy}}(t_j))\}_{j=1}^{N_{\text{BC}}}.$$

3. Interior collocation points

$$\mathcal{D}_f^{(s)} = \{(x_{f,j}, t_{f,j})\}_{j=1}^{N_f^{(s)}},$$

sampled in $\Omega_s \times (0, T]$ using Latin hypercube sampling.

4. Interface points

$$\mathcal{D}_{\text{int}}^{(s,s+1)} = \{(x_{s+1/2}, t_j)\}_{j=1}^{N_{\text{int}}},$$

sampled along each shared subdomain boundary.

5. A structured regularization grid

$$\mathcal{D}_{\text{ext}}^{(s)} = \{(x_i, t_j)\},$$

used only by the extremum-regularized model. The points are uniformly ordered in x at every fixed time so that the neighboring predictions U_{i-1} , U_i , and U_{i+1} are well defined.

The numbers of points used in the experiments are

$$N_{\text{IC}}^{(s)} = 200, \quad N_{\text{BC}} = [\text{insert value}], \quad (34)$$

$$N_f^{(s)} = 4000, \quad N_{\text{int}} = 300. \quad (35)$$

The structured regularization grid contains

$$N_x^{\text{ext}} = 100$$

spatial points and

$$N_t^{\text{ext}} = 20$$

time levels per subdomain.

Initial-condition labels are evaluated directly from Equation (16). If boundary trajectories are taken from the finite-volume reference rather than prescribed analytically, this must be stated explicitly, and the neural experiment must be described as reference-assisted.

2.6. Baseline cPINN Loss

The baseline loss in subdomain Ω_s is

$$\mathcal{L}_{\text{base}}^{(s)} = w_{\text{data}} \mathcal{L}_{\text{data}}^{(s)} + w_{\text{PDE}} \mathcal{L}_{\text{PDE}}^{(s)} + w_{\text{int}} \mathcal{L}_{\text{int}}^{(s)}. \quad (36)$$

2.6.1 Initial- and Boundary-Data Loss

The data loss is

$$\mathcal{L}_{\text{data}}^{(s)} = \mathcal{L}_{\text{IC}}^{(s)} + \mathcal{L}_{\text{BC}}^{(s)}, \quad (37)$$

where the initial-condition contribution is

$$\mathcal{L}_{\text{IC}}^{(s)} = \frac{1}{N_{\text{IC}}^{(s)}} \sum_{j=1}^{N_{\text{IC}}^{(s)}} \|U_{\theta_s}(x_j, 0) - U_0(x_j)\|_2^2. \quad (38)$$

The boundary contribution is defined analogously on the external domain boundaries.

2.6.2 PDE-Residual Loss

The inviscid isentropic Euler residuals are

$$r_\rho = \partial_t \rho + \partial_x m, \quad (39)$$

$$r_m = \partial_t m + \partial_x \left(\frac{m^2}{\rho} + K \rho^\gamma \right). \quad (40)$$

In the reported neural experiments, the training residual includes an artificial-viscosity coefficient ε_{av} :

$$r_\rho^{\text{av}} = \partial_t \rho + \partial_x m - \varepsilon_{\text{av}} \partial_{xx} \rho, \quad (41)$$

$$r_m^{\text{av}} = \partial_t m + \partial_x \left(\frac{m^2}{\rho} + K \rho^\gamma \right) - \varepsilon_{\text{av}} \partial_{xx} m. \quad (42)$$

All derivatives are evaluated by automatic differentiation. The PDE loss is

$$\mathcal{L}_{\text{PDE}}^{(s)} = \frac{1}{N_f^{(s)}} \sum_{j=1}^{N_f^{(s)}} \left[|r_{\rho,j}^{\text{av}}|^2 + |r_{m,j}^{\text{av}}|^2 \right]. \quad (43)$$

The artificial-viscosity coefficient is

$$\varepsilon_{\text{av}} = 2 \times 10^{-3}.$$

The same coefficient is used in the baseline and regularized cPINNs so that the only difference between them is the additional extremum penalty. Because $\varepsilon_{\text{av}} > 0$ introduces diffusion into the training residual, the neural model is trained using a viscously regularized approximation of the inviscid equations [5], [6].

2.6.3 Interface Loss

At an interface $x = x_{s+1/2}$ between subdomains Ω_s and Ω_{s+1} , the state-matching contribution is

$$\mathcal{L}_{\text{state}}^{(s,s+1)} = \frac{1}{N_{\text{int}}} \sum_{j=1}^{N_{\text{int}}} \|U_{\theta_s}(x_{s+1/2}, t_j) - U_{\theta_{s+1}}(x_{s+1/2}, t_j)\|_2^2. \quad (44)$$

The physical-flux-matching contribution is

$$\mathcal{L}_{\text{flux}}^{(s,s+1)} = \frac{1}{N_{\text{int}}} \sum_{j=1}^{N_{\text{int}}} \|F(U_{\theta_s}(x_{s+1/2}, t_j)) - F(U_{\theta_{s+1}}(x_{s+1/2}, t_j))\|_2^2. \quad (45)$$

The interface loss is

$$\mathcal{L}_{\text{int}}^{(s,s+1)} = \mathcal{L}_{\text{state}}^{(s,s+1)} + \mathcal{L}_{\text{flux}}^{(s,s+1)}. \quad (46)$$

This interface loss is present in both neural models and arises from the cPINN domain decomposition, not from the extremum regularizer.

2.7. Minmod-Inspired Extremum Regularization

For the regularized cPINN, the network is evaluated on the structured grid described in Section 2.5. For a scalar predicted quantity q_i , where q is either ρ or m , define

$$\Delta_i^- q = q_i - q_{i-1}, \quad \Delta_i^+ q = q_{i+1} - q_i. \quad (47)$$

The local penalty is

$$P_i(q) = \text{ReLU}(-\Delta_i^- q \Delta_i^+ q). \quad (48)$$

This term is positive when

$$\Delta_i^- q \Delta_i^+ q < 0,$$

which identifies a strict discrete local maximum or minimum. It is zero on locally monotone data. It can also be zero at an exactly monotone discontinuity when one neighboring difference is zero.

The penalty does not determine whether an extremum is physical or nonphysical. In particular, it can activate at both an artificial oscillation and a legitimate smooth extremum. It should therefore be interpreted as an extremum-suppression regularizer rather than as a shock detector.

The subdomain regularization loss is

$$\mathcal{L}_{\text{ext}}^{(s)} = \frac{1}{N_{\text{ext}}^{(s)}} \sum_{(i,j) \in \mathcal{D}_{\text{ext}}^{(s)}} [P_{i,j}(\rho) + P_{i,j}(m)]. \quad (49)$$

The complete regularized objective is

$$\mathcal{L}_{\text{reg}}^{(s)} = \mathcal{L}_{\text{base}}^{(s)} + \lambda_{\text{ext}} \mathcal{L}_{\text{ext}}^{(s)}. \quad (50)$$

Setting

$$\lambda_{\text{ext}} = 0$$

recovers the baseline cPINN. The preliminary comparison uses

$$\lambda_{\text{ext}} = 5.$$

This value is treated as an initial test value rather than an optimized choice. The planned sensitivity study evaluates

$$\lambda_{\text{ext}} \in \{0, 0.1, 1, 5\}$$

using common random seeds.

Unlike the classical minmod limiter in Equation (19), Equation (48) does not alter a reconstructed state or a conservative numerical flux. The two mechanisms are therefore not mathematically equivalent.

2.8. Loss Weights and Optimization

The loss weights are

$$w_{\text{data}} = 10, \quad w_{\text{PDE}} = 5, \quad w_{\text{int}} = 10. \quad (51)$$

Both neural models use the same architecture, training points, loss weights, artificial-viscosity coefficient, initialization procedure, and optimization budget. This controlled setup ensures that the extremum penalty is the only intended difference between the models.

Training is performed in two stages.

2.8.1 Adam Stage

The networks are first trained using Adam [7] for

$$N_{\text{Adam}} = 20,000$$

iterations with initial learning rate

$$\eta_0 = 1 \times 10^{-3}.$$

The learning-rate schedule is

The learning-rate schedule is exponential decay, $\eta_{it} = \eta_0 \cdot 0.9^{\lfloor it/2000 \rfloor}$, decaying by a factor of 0.9 every 2000 iterations.

The scattered interior and interface points are resampled every

$$R = 2000$$

iterations. The structured extremum grid is not resampled because its ordered spatial points are required to calculate neighboring differences.

2.8.2 L-BFGS Stage

After Adam training, all subdomain parameters are optimized jointly using L-BFGS-B for

$$N_{\text{L-BFGS}} = 4000$$

iterations, with stopping tolerance

$$ftol = 1 \times 10^{-12}, gtol = 1 \times 10^{-10}.$$

The experiments use the random seeds 1234 (set via `np.random.seed(1234)` and `set_random_seed(1234)`). The hardware, software framework, numerical precision, and approximate training time are reported through an M5 CPU and using the `tf.compat.v1` library under `tf.disable_v2_behavior` and `float64` precision.

2.9. Evaluation Metrics

At the final time T , the four subdomain predictions are evaluated on a common fine grid and concatenated to form a global prediction over $[0, 1]$. Values at shared interfaces are assigned from each subdomain's own network restricted strictly to its own bounds $[x_s^L, x_s^R]$, with no averaging between adjacent subdomain predictions at the shared boundary..

For a predicted scalar quantity q_θ and reference values q_{ref} , the relative discrete L^1 error is

$$E_{L^1}(q) = \frac{\sum_{i=1}^{N_{\text{eval}}} |q_{\theta,i} - q_{\text{ref},i}|}{\sum_{i=1}^{N_{\text{eval}}} |q_{\text{ref},i}| + \epsilon_{\text{den}}}, \quad (52)$$

and the relative discrete L^2 error is

$$E_{L^2}(q) = \frac{\left[\sum_{i=1}^{N_{\text{eval}}} (q_{\theta,i} - q_{\text{ref},i})^2 \right]^{1/2}}{\left[\sum_{i=1}^{N_{\text{eval}}} q_{\text{ref},i}^2 \right]^{1/2} + \epsilon_{\text{den}}}. \quad (53)$$

Here, ϵ_{den} is a small constant used only to prevent division by zero.

The total variation of density is

$$TV(\rho_\theta) = \sum_{i=1}^{N_{\text{eval}}-1} |\rho_{\theta,i+1} - \rho_{\theta,i}|. \quad (54)$$

Density overshoot and undershoot relative to the reference range are measured by

$$E_{\text{over}} = \max \left(0, \max_i \rho_{\theta,i} - \max_i \rho_{\text{ref},i} \right), \quad (55)$$

$$E_{\text{under}} = \max \left(0, \min_i \rho_{\text{ref},i} - \min_i \rho_{\theta,i} \right). \quad (56)$$

The estimated shock-location error is

$$E_{\text{shock}} = \left| x_{\text{shock}}^{\text{pred}} - x_{\text{shock}}^{\text{ref}} \right|, \quad (57)$$

where the shock position is estimated from

$$x_{\text{shock}} = \arg \max_{x_i} \left| \frac{\rho_{i+1} - \rho_{i-1}}{2\Delta x} \right|.$$

The regularization loss $\mathcal{L}_{\text{ext}}$ is also recorded during training. A decrease in this loss indicates that the network is producing fewer or weaker strict discrete extrema on the structured grid. It does not, by itself, prove improved accuracy, conservation, shock resolution, or physical admissibility.

2.10. Controlled Preliminary Comparison

The preliminary comparison contains two neural models:

1. the baseline cPINN with $\lambda_{\text{ext}} = 0$;
2. the extremum-regularized cPINN with $\lambda_{\text{ext}} = 5$.

Both models use identical network architectures, training data, subdomain boundaries, artificial viscosity, optimizer settings, evaluation grids, and random seeds. The current results are treated as preliminary single-configuration results. They demonstrate whether the cPINN captures the principal

wave structure of the finite-volume reference, but they do not establish statistical superiority or a numerical convergence rate.

The complete study will repeat the experiment for

$$\lambda_{\text{ext}} \in \{0, 0.1, 1, 5\}$$

and for the common seeds

$$\{11, 22, 33\}.$$

For each configuration, the final manuscript will report the mean and standard deviation of the relative L^1 and L^2 errors, total variation, overshoot, undershoot, shock-location error, and training time.

3. PRELIMINARY RESULTS AND DISCUSSION

This section presents the currently available single-run results. The purpose of this preliminary comparison is to determine whether the baseline and extremum-regularized cPINNs reproduce the principal wave structure of the numerical reference solution. Because the neural models have not yet been evaluated over multiple random initializations or penalty weights, the reported errors should not be interpreted as statistically significant comparisons or as evidence of numerical convergence.

3.1. Baseline cPINN

Figure 1 compares the density predicted by the baseline cPINN with the finite-volume reference solution at the final time $t = 0.150$. The cPINN prediction reproduces the principal features of the reference profile, including the left constant state, the rarefaction region, the intermediate state, and the sharp transition on the right.

The visual agreement provides preliminary evidence that the domain-decomposed neural model captures the dominant wave structure. However, because only one network architecture, training budget, collocation-point configuration, and random initialization are shown, this result should be described as qualitative agreement rather than as proof of convergence.

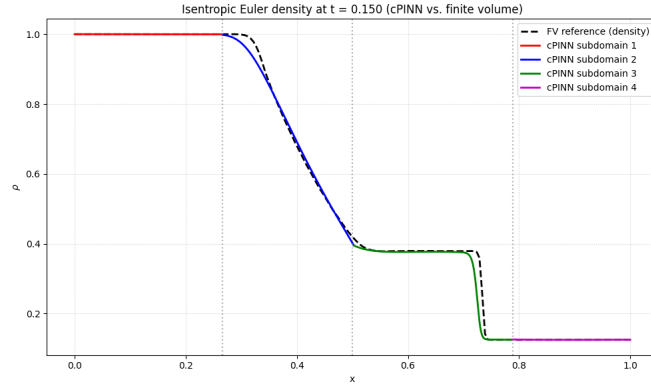


Figure 1. Density predicted by the baseline cPINN at $t = 0.150$, compared with the finite-volume reference solution. The four colored curves represent the predictions from the four cPINN subdomains. The model reproduces the principal wave structure, although differences remain near the sharp transition.

3.2. Extremum-Regularized cPINN

Figure 2 shows the corresponding prediction from the extremum-regularized cPINN. The regularized model also reproduces the main wave structure of the finite-volume reference. Its profile is visually similar to that of the baseline cPINN.

For the reported run, the relative L^2 density error of the extremum-regularized cPINN is

$$E_{L^2, \text{reg}} = 2.6859 \times 10^{-2}, \quad (58)$$

whereas the relative L^2 density error of the baseline cPINN is

$$E_{L^2, \text{base}} = 1.9281 \times 10^{-2}. \quad (59)$$

Therefore, the regularized model has the larger error in this preliminary comparison. Relative to the baseline value, the increase is approximately

$$\frac{E_{L^2, \text{reg}} - E_{L^2, \text{base}}}{E_{L^2, \text{base}}} \times 100\% \approx 39.3\%. \quad (60)$$

This result does not establish that the extremum regularizer is generally detrimental, because only one training configuration is considered and neural-network optimization can depend on initialization and collocation sampling. Nevertheless, the result does show that adding the penalty with $\lambda_{\text{ext}} = 5$ does not improve the error in the presently reported run. Consequently, $\lambda_{\text{ext}} = 5$ should be described as a preliminary test value rather than an optimal choice.

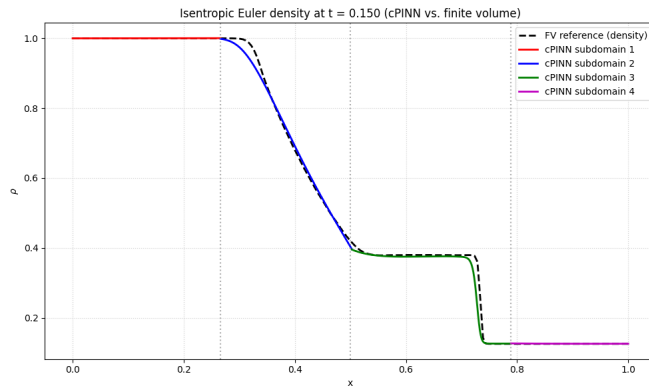


Figure 2. Density predicted by the extremum-regularized cPINN at $t = 0.150$, compared with the finite-volume reference solution. The regularized model captures the principal wave structure, but its relative L^2 density error is larger than that of the baseline cPINN in the reported run.

The two neural results illustrate an important distinction between qualitative and quantitative agreement. Both models capture the visible structure of the reference solution, but visual similarity alone does not imply equal accuracy. For the available run, the quantitative error favors the baseline cPINN.

3.3. Finite-Difference Comparison

Figure 3 compares the density profiles produced by the two displayed finite-difference configurations with the reference solution. Both numerical profiles reproduce the overall wave pattern but visibly smear the rarefaction and sharp transition relative to the reference curve.

The figure reports an L^2 error of approximately

$$4.30 \times 10^{-2}$$

for both displayed numerical solutions. Therefore, the present figure does not demonstrate a measurable error advantage for one displayed numerical scheme over the other. The two numerical curves are also nearly indistinguishable at the plotting resolution.

The similar global errors may result from the error being concentrated near the nonsmooth parts of the solution. Near a shock, a limiter commonly reduces oscillations by lowering the effective local reconstruction order and introducing additional numerical diffusion [8]. In the present experiment, however, this explanation should be regarded as a plausible interpretation rather than a demonstrated mechanism because shock width, overshoot, and local error have not been measured separately.

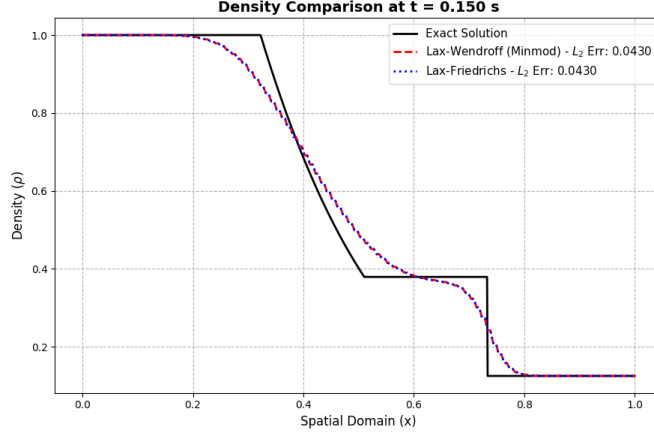


Figure 3. Density profiles from the two displayed finite-difference configurations at $t = 0.150$, compared with the reference solution. The legend reports an L^2 error of approximately 4.30×10^{-2} for both numerical profiles. The exact names in this caption and in the figure legend must be made consistent with the algorithms actually implemented in the code.

The figure legend currently identifies the displayed schemes as “Lax–Wendroff (Minmod)” and “Lax–Friedrichs.” These names must be checked against the implementation. If the limited method actually uses MUSCL reconstruction and a Rusanov flux, it should be labeled “MUSCL–minmod–Rusanov,” not “Lax–Wendroff (Minmod).” A method name should describe the algorithm that was executed rather than combining labels from different discretizations.

3.4. Interpretation of the Preliminary Findings

The currently available results support three limited conclusions:

1. Both cPINN models reproduce the principal wave structure of the finite-volume reference solution.
2. In the reported single run, the baseline cPINN has a smaller relative L^2 density error than the extremum-regularized cPINN.
3. The value $\lambda_{\text{ext}} = 5$ has not been shown to be optimal, and the current results do not demonstrate that the extremum penalty improves accuracy.

These findings are consistent with the mathematical distinction established earlier in the paper. A classical minmod limiter modifies reconstructed interface states before a conservative numerical flux is calculated. The neural extremum penalty modifies only the training objective. It may reduce strict discrete extrema without necessarily improving the location or sharpness of a shock.

4. CONCLUSION AND FUTURE WORK

This study examined a minmod-inspired local-extremum regularizer for a conservative physics-informed neural network applied to the one-dimensional isentropic Euler equations. The proposed term penalizes sign changes between neighboring discrete differences of the neural prediction. Unlike a classical finite-volume flux limiter, it does not reconstruct interface states or modify a conservative numerical flux. It should therefore be interpreted as an extremum-suppression regularizer rather than as a flux limiter.

The preliminary results show that both the baseline and regularized cPINNs reproduce the principal wave structure of the finite-volume reference solution. This qualitative agreement demonstrates that the implemented cPINN can approximate the dominant features of the considered Riemann problem. In the currently reported run, however, the regularized model has a relative L^2 density error of

2.6859×10^{-2} , compared with 1.9281×10^{-2} for the baseline model. Thus, the addition of the extremum penalty with $\lambda_{\text{ext}} = 5$ does not improve the reported error.

The result illustrates that suppressing discrete extrema is not equivalent to improving solution accuracy. A regularization term can alter the optimization process and smooth parts of the predicted solution without correcting shock location, shock width, or conservation error. The present experiment therefore motivates careful distinction between flux-based stabilization and loss-based neural regularization. The primary limitations of the current study are that it considers one Riemann problem and reports one preliminary neural-network run for each model. The results do not establish a convergence rate, statistical superiority, total-variation-diminishing behavior, or entropy stability. In addition, the effect of the extremum penalty is considered together with the artificial-viscosity term already present in the neural residual.

Future work will examine the sensitivity of the result to the regularization weight, random initialization, and structured-grid resolution. It may also investigate regularization that is activated by a more selective shock sensor. A longer-term extension would place a genuine limited reconstruction inside a control-volume or numerical-flux residual so that the limiter acts on conservative interface fluxes rather than only on the neural training loss. The principal conclusion of the present preliminary study is that the cPINN can qualitatively reproduce the finite-volume wave structure, but the tested minmod-inspired extremum penalty does not improve the reported single-run error. This negative result is scientifically useful because it clarifies that importing the algebraic form of a limiter into a neural loss does not reproduce the mechanism of classical flux limiting.

5. ACKNOWLEDGMENTS

In this manuscript, the reported neural-network comparisons are based on single training runs. Sensitivity to random initialization, regularization weight, sampling, and numerical resolution has not yet been evaluated. I would like to thank Dr. Sarswati Shah for all her help in working with me with classical numerical schemes.

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